

UTKARSH SARVJEET TIWARI

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Experience

NLIP Lab, Indian Institute of Technology Hyderabad

Sangareddy, India

Research Intern (AI/ML)

Feb 2026 – Present

- Contributing to development of an AI-based product/system.
- Building data pipelines and preprocessing workflows for real-world datasets.
- Performing model evaluation and optimization to improve accuracy and response time.
- Collaborating with research scholars and faculty on system architecture and experimentation.

Aryahs World Infotech (OPC) Pvt. Ltd.

Navi Mumbai, Maharashtra

AI/ML Intern

Dec 2025 – Jan 2026

- Developed WhatsApp automation system for customer interaction and workflow management.
- Built ML-based symptom classifier mapping user inputs to probable medical conditions.
- Implemented automated email generation system for context-aware communication.
- Contributed to full-stack product features using Python, Flask, MySQL & JavaScript.
- Worked with real-time datasets, testing, and Agile sprint collaboration.

Cybranex Pvt. Ltd. (Industry-Defined Research, under Faculty Mentorship)

Remote

Quantum Computing Researcher

Oct 2024 – Present

- Designing and training quantum autoencoder models for data compression and feature extraction.
- Simulating quantum circuits and exploring hybrid quantum–classical architectures for dimensionality reduction.
- Conducting research and documentation as part of a college–industry collaboration program.

Education

Shah and Anchor Kutchhi Engineering College

Mumbai, Maharashtra

B.Tech in Electronics and Computer Science Engineering — CGPA: 8.4/10.0

Sept 2023 – 2027 (Expected)

- **Certifications:** Linux Essentials – Cisco Networking Academy; Embedded System Developer – AICTE Virtual Internship (Eduskills Foundation).
- **Leadership:** Led 12 campus ambassadors at ISRC; Finalist in multiple national AI hackathons.

Skills

Languages: Python, C, C++, JavaScript

AI & Machine Learning: Machine Learning, NLP, Sentence Transformers, TensorFlow, Keras, Scikit-learn

Data & Visualization: NumPy, Pandas, Matplotlib, Data Preprocessing, Model Evaluation

Web & Frameworks: Flask, Node.js, React.js (JSX)

Databases & Tools: MySQL, MongoDB, Git, VS Code, XAMPP, OpenCV

Quantum (Basics): Qiskit, PennyLane

Projects

Local Voice AI Agent | Python, Ollama, Gradio, WebRTC, YAML | [GitHub](#)

Aug 2025

- Developed a Windows-first voice chat app using local AI models (Gemma 3 via Ollama) with real-time STT/TTS pipeline.
- Implemented conversational memory, retry logic, and performance metrics while defaulting to 1B model for minimal downloads.
- Created configurable CLI with tuning parameters and multiple deployment modes (local/LAN/public sharing).

Ecommerce Web Application – Artha Mart | Flask, MySQL, Python, JavaScript | [GitHub](#)

Jun 2025

- Developed a group-based e-commerce web application providing a seamless online shopping experience.
- Designed and implemented MySQL database schema for users, products, orders, and cart management.
- Built backend logic in Flask to handle user authentication, product management, cart operations, and order processing.
- Wrote secure and efficient SQL queries for data retrieval and manipulation.

Harry Potter Invisibility Cloak | Python, OpenCV, NumPy | [GitHub](#)

Mar 2025

- Developed a computer vision project simulating the invisibility effect using color detection and HSV masking in real-time video.
- Utilized OpenCV and NumPy libraries to capture background, apply color masks, and blend frames to produce the invisibility illusion.
- Included real-time trackbar tuning and a demo video demonstration.

Smart Home Automation System | C++, File I/O | [GitHub](#)

Dec 2024

- Device control & energy monitoring simulator for smart home automation scenarios.

Awards & Hackathon Experience

- **Innovate-a-thon, IIT (ISM) Dhanbad (Concetto'25)** – Finalist; proposed *Reloop*, an AI-driven circular economy platform.
- **HackWithUttarPradesh 2025** – Shortlisted; Smart Irrigation System using IoT sensors and ML.
- **HackSetu 1.0, Amity University Gwalior** – Grand Finale; AI + IoT irrigation prototype with real-time sensor data.
- **E² Calomics'25** – Paper Presentation Selection; ML approach for optimizing irrigation using environmental parameters.
- **Bharatiya Antariksh Hackathon 2025** – Participant; AI-based drought and land risk prediction using satellite imagery.